

WAYNE, NE, USA

WMO: 722241

Lat: 42.242N

Lon: 96.983W

Elev: 434

StdP: 96.22

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-18.6	-25.1	0.4	-21.1	-22.5	0.5	-18.0	14.3	-3.1	12.8	-4.7	2.4	320	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	33.9	22.7	32.2	22.5	30.6	22.0	25.8	29.8	24.9	29.2	23.9	28.3	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	21.0	28.4	23.6	19.4	27.4	22.5	18.1	26.6	82.7	29.8	78.0	29.3	73.8	28.2	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	11.1	9.7	DB	-27.8	36.2	3.1	1.8	-30.0	37.4	-31.8	38.5	-33.5	39.5	-35.7	40.7	
			WB	-28.0	27.7	3.0	1.2	-30.1	28.6	-31.9	29.3	-33.6	30.0	-35.8	30.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-5.9	-4.8	3.2	8.8	15.4	21.4	23.2	21.5	17.7	9.8	2.7	-4.4
	DBStd	11.70	6.94	6.87	6.70	5.54	5.07	3.63	3.37	2.98	4.59	5.33	5.77	6.26
	HDD10.0	1974	494	416	227	85	10	0	0	0	3	68	225	446
	HDD18.3	3807	752	649	469	288	119	12	4	7	66	269	468	704
	CDD10.0	1658	0	1	18	50	179	342	409	356	233	63	7	0
	CDD18.3	449	0	0	1	3	29	104	155	105	47	6	0	0
	CDH23.3	4773	0	0	20	73	393	1121	1657	930	502	76	2	0
	CDH26.7	1728	0	0	3	23	142	410	671	297	157	26	0	0
(14) Wind	WSAvg	4.1	4.3	4.5	4.8	5.1	4.7	3.9	3.0	2.8	3.5	4.0	4.2	4.1
(15) Precipitation	PrecAvg	717	17	21	53	74	103	112	83	72	73	52	38	20
	PrecMax	1034	51	86	167	285	232	315	261	201	244	160	114	77
	PrecMin	413	1	1	4	11	32	25	9	12	7	1	0	1
	PrecStd	148	11	18	40	50	43	58	50	44	46	38	31	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	16.7	26.6	30.1	33.4	35.2	37.1	34.9	33.2	30.3	22.4	12.9
		MCWB	6.7	9.2	15.1	16.0	19.2	22.5	23.0	22.8	21.3	19.3	13.0	7.4
	2%	DB	8.1	11.0	21.8	25.1	30.7	32.7	34.5	31.7	30.5	25.0	18.5	9.9
		MCWB	4.0	6.3	12.2	13.7	18.6	22.2	23.6	23.6	21.3	15.5	11.2	5.8
	5%	DB	5.1	7.3	17.6	21.8	27.6	31.0	32.3	30.0	28.2	22.3	15.0	6.3
		MCWB	2.5	4.0	11.8	12.9	17.3	21.2	24.3	23.0	20.1	14.6	8.9	3.3
	10%	DB	2.8	4.3	13.8	19.0	25.0	29.0	30.3	27.9	26.3	19.4	12.0	3.6
		MCWB	1.2	2.3	9.1	11.5	16.6	20.6	23.3	22.1	19.3	13.2	7.5	1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	10.6	16.9	17.7	22.4	26.2	28.0	26.4	24.9	20.6	14.6	9.0
		MCDB	13.0	17.5	22.5	25.1	28.1	30.2	31.1	30.2	28.6	26.3	19.9	11.5
	2%	WB	4.5	6.4	14.5	15.8	20.7	24.6	26.5	25.3	23.3	17.9	11.9	5.6
		MCDB	7.5	10.4	18.9	22.0	26.7	29.5	30.2	29.1	27.5	22.8	17.5	8.6
	5%	WB	2.8	4.3	12.0	14.2	19.1	23.3	25.3	24.3	22.1	15.9	9.8	3.6
		MCDB	4.9	7.1	17.3	20.1	24.8	28.2	29.6	28.1	26.3	20.1	14.5	6.4
	10%	WB	1.3	2.3	9.4	12.5	17.6	22.3	24.2	23.2	20.8	13.6	7.6	1.9
		MCDB	2.9	4.3	14.6	17.8	23.4	27.0	28.9	27.2	24.8	18.8	11.8	3.6
(35) Mean Daily Temperature Range	MDBR		10.4	10.6	11.9	13.2	13.1	12.4	12.3	11.7	13.8	13.6	12.8	9.9
	5% DB	MCDBR	13.0	14.2	18.8	19.0	16.9	14.3	13.8	13.8	15.4	18.0	17.8	13.5
		MCWBR	9.2	10.0	10.7	9.6	7.5	6.0	6.0	7.0	7.2	9.4	10.6	9.9
	5% WB	MCDBR	12.3	13.6	16.7	15.2	13.5	11.5	11.7	11.5	11.9	14.1	16.4	12.3
		MCWBR	9.0	9.9	10.0	8.6	7.1	6.4	6.7	7.0	7.0	8.5	10.6	9.3
(40) Clear-Sky Solar Irradiance	taub		0.274	0.290	0.317	0.389	0.411	0.418	0.424	0.421	0.371	0.323	0.287	0.278
	taud		2.448	2.436	2.417	2.223	2.237	2.278	2.268	2.270	2.373	2.493	2.507	2.454
	Ebn at Noon		882	922	931	878	865	857	847	838	860	871	860	844
	Edh at Noon		74	89	103	135	137	132	132	128	107	84	70	68
(44) All-Sky Solar Radiation	RadAvg		1.97	2.98	3.86	4.93	5.78	6.48	6.62	5.59	4.53	3.15	2.20	1.62
	RadStd		0.17	0.23	0.31	0.35	0.42	0.47	0.34	0.41	0.31	0.37	0.15	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page